

02_HETEROGENEOUS_EDGE_MESH_AND_FLYWHEEL.md

Heterogeneous Edge-Mesh Architecture, Guardrails & The Recursive Flywheel

1. The Network Layer: Heterogeneous P2P Edge-Mesh

- **Nodes:** Mobile Engine (Node Alpha: Snapdragon 8 Elite), Compute Server (Node Beta: NVIDIA Jetson Orin Nano Super), and Auxiliary Dev Boards (Rubik Pi 3).
 - **Operation:** Ad-hoc local peer-to-peer Wi-Fi/Tailscale mesh eliminates cloud latency and prevents data egress. Node Alpha functions as the ambient sensory and triage ingress, offloading heavy token processing and compilation tasks seamlessly to Node Beta.
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2. Multi-Tier Programmatic Cognitive Memory Engine (CME)

Memory is dynamic and graph-relational across split services:

- **mem0:** Short-term episodic caching tracking shifting conversational facts and user formatting habits.
 - **OB1 (Open Brain):** Ground-truth vector persistence hosted on PostgreSQL with pgvector for stable multi-turn retrieval over MCP.
 - **OmniRoute Gateway:** Central data router on port 20128 providing RTK token compression and an asynchronous memory extraction tap into the Reasoning Bank.
 - **Graphify & notebooklm-py:** Converts flat execution logs into visual code-review graphs and drives headless multi-document analytical research pipelines.
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3. Out-of-Band Security & Adversarial Gatekeeping

- **The Sealed Red Auditor:** Operates within an air-gapped sandbox (`.incognito_red_sandbox/`).
 - **Quarantine Protocol:** Intercepts candidate batch traces, validating JSON syntax and cross-referencing against negative assertion banks to strip hallucinated tool calls or system exploits before any file touches permanent storage.
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4. Recursive Knowledge-Augmented Generation (KAG) Flywheel

- Audited and verified execution traces are serialized into `.jsonl` formats and archived to cloud training buckets (`gs://novae-rlvr/`).
- The system utilizes verified historical performance to drive continuous reinforcement learning, fine-tuning local model weights and refining procedural prompts online without requiring manual human re-scaffolding.